

# Codeforces Round 1075 — Problems C1 and C2

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*The problem statement is worded differently here.*

## Codeforces Round 1075 — Problems C1 and C2

Given a natural number  $n$ . Find a permutation  $p$  of  $(1, 2, \dots, n)$ , such that for each  $i \in S$  there exists a  $i \leq j \leq n$  such that  $p_i = p_j \oplus i$ .

C1 — Solve the problem for  $S = \{2, 3, \dots, n-1\}$ .

C2 — Solve the problem for  $S = \{1, 2, \dots, n-1\}$ .

### Input

Each test contains multiple test cases. The first line contains the number of test cases  $1 \leq t \leq 10^4$ .

The description of the test cases follows. The only line of each test case contains one positive integer  $3 \leq n \leq 2 \cdot 10^5$ , the length of the permutation.

It is guaranteed that the sum of  $n$  over all test cases does not exceed  $2 \cdot 10^5$ .

### Output

If there is a suitable permutation output the  $n$  integers  $p_1 p_2 \dots p_n$  with spaces between them. Otherwise output  $-1$ .

## Solution

*If we leave out any details in the solution then it's because, either it shouldn't be too difficult to fill in the gaps.*

C1 — We look for an easy way to satisfy the desired condition. After some exploration one may notice that

$$a = \begin{cases} (a-1) \oplus 1 & \text{if } a \text{ is odd,} \\ (a+1) \oplus 1 & \text{if } a \text{ is even.} \end{cases}$$

This is the heart of the problem. The claim is true because if  $a$  is odd then  $(a-1)_2$  is just  $(a)_2$  without the  $2^0$  bit. Now if you XOR with 1, you are adding the  $2^0$  bit back in, giving  $a = (a-1) \oplus 1$ . Similarly, if  $a$  is even then  $(a+1)_2$  is just  $(a)_2$  with a  $2^0$  bit added.\* Thus if one XOR's with 1 then you will remove this  $2^0$  bit again giving  $a = (a+1) \oplus 1$ .

So if we place 1 at the end of our permutation and let

$$p_i = \begin{cases} i+1 & \text{if } i \text{ is even,} \\ i-1 & \text{if } i \text{ is odd.} \end{cases}$$

then just let  $p_1$  take on the one unused value, then we can take  $j = n$  to get  $p_n \oplus i = p_i$ , and this can be checked to work for all suitable  $2 \leq i \leq n-1$ .

\* Note that  $(a)_2$  didn't have this bit before because only odd numbers have this bit in their binary representation.

C2 — One can check that if  $n$  is odd then our construction from C1 works fine. The only issue is with even  $n$ . Again after some experimentation one may find that the construction is never possible if  $n$  is a power of 2.

This is true because if  $a \oplus b = 2^k$  for  $1 \leq b \leq a \leq 2^k$  then  $a$  must have the  $2^k$  bit in its binary representation. Now we notice that whatever bits  $b$  has (other than the  $2^k$  bit),  $a$  must also have these bits to satisfy the XOR equation. Since  $b \geq 1$  we have that  $a \geq 2^k + 1$  a contradiction.

Now if  $n$  is even but isn't a power of 2, then we look for a construction that works in this occasion. Let's start out from our construction in C1. If you write out some cases, you'll find that the only issue occurs when  $i = 1$  and we want to find  $j$  such that  $p_1 = n = p_j \oplus 1$ . It is natural then to guess that perhaps we can use our construction from C1, except just swap  $p_1$  and some other number. Let  $l \geq 0$  be an integer such that  $2^l$  is the smallest bit that occurs in the binary representation of  $n$ . If you swap the elements at  $p_1$  and  $p_{2^l}$ , then the resultant permutation satisfies the desired criteria. *We leave the proof of this fact as an exercise to the reader, as a hint look at the number  $n - 2^l$ .* ■